

Mapping Displacement and Gentrification: A Machine Learning Approach to Brooklyn's Urban Change

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Introduction

Gentrification has been a contested issue in urban studies since the term was first coined. It is a pervasive phenomenon observed interdisciplinarily in cities worldwide, and often blamed for being the impetus of urban injustice and displacement (Lees, 2015). In the great book *The New Urban Frontier*, Neil Smith (1996: p32) stressed that gentrification is “indeed a dirty word and it would stay a dirty word,” as it is deeply connected with the violent process of displacement, where people become impoverished, get evicted, or fall into homelessness. However, some scholars doubt that the interconnectedness between gentrification and displacement lacks empirical evidence. Recent research, like Freeman & Braconi (2004), used statistical models to measure the relationship between gentrification and displacement, did not find strong links between gentrification and displacement. Others admit that gentrification could induce displacement, which is represented by the changes in household size, racial composition, annual income, age, and education level, but insist it has benefits that outweigh the negative outcomes (Gale, 2007).

Not only is gentrification a “chaotic concept” that can seldom offer specific and meaningful policy implications (Rose, 1984), but displacement, a concept often associated with gentrification, is also in need of clarification, both theoretically and empirically. On many occasions, urban displacement is considered to be linked to gentrification, whether directly or indirectly (Marcuse,

1985). However, this approach is challenged by scholars who argue that displacement could take many forms other than physical ones, and the causes of displacement could be many, depending on the local context (Watt & Morris, 2024; Van, 2025).

This project aims to apply machine learning techniques to group areas by their state of displacement and explore the degree to which gentrification induced displacement. We combine multiple supervised learning and unsupervised learning methods to build interpretable and robust models, while also using them to predict hypothesized scenarios. We chose Brooklyn, New York, as our major geographical focus to exemplify our approach. By leveraging machine learning techniques, we aim to address the methodological gap in gentrification and displacement research and contribute to both theoretical and practical discussions.

Literature Review

Gentrification has been examined through various theoretical lenses, both qualitatively and quantitatively. Qualitative research, like Smith's (1996) article "Class Struggle on Avenue B", often engages with protests and uprisings in certain gentrifying neighborhoods. This approach implies a lot on the underlying mechanisms behind displacement, and provide theoretical foundations for quantitative research. Quantitative research often utilizes socio-economic data to observe the patterns of gentrification across neighborhoods and cities, which allows a better understanding of the scope and scale of gentrification.

Quantitative research on gentrification uses various kinds of data to identify patterns. Longitudinal statistical analysis on American Housing Survey data along with American Community Survey data are often used to research the patterns of gentrification in US cities, with

considerations of demographic data (age, gender, race), housing characteristics, income, education, amenity, marriage status, and family structure (Yoo, 2023; Freeman & Braconi, 2004; Newman & Wyly, 2006). Chapple et al. (2021) explored using geotagged Twitter data in combination with census data to enhance the understanding of neighborhood changes in the San Francisco Bay Area. Chapple et al. (2021) also established the *Urban Displacement Project*, which identifies gentrification and displacement typologies in metropolitan areas worldwide based on census data.

Other research goes further to predict displacement with machine learning. Alejandro and Palafox (2019) apply machine learning methods, specifically Random Forest Classifiers, to predict neighborhood-level gentrification in Mexico City. Similarly, Reades et al. (2019) utilized Random Forest to analyze and predict patterns of socioeconomic change in London neighborhoods between 2001 and 2021. They integrate machine learning with qualitative approaches to justify the use of supervised learning for neighborhood classification and inform the incorporation of explainability tools like SHAP (SHapley Additive exPlanations). Mueller et al. (2024) further synthesized emerging machine learning approaches in gentrification studies, highlighting how clustering, computer vision, and explainable AI can overcome challenges of traditional statistical models. Their emphasis on transparency and interpretability supports the responsible application of predictive tools in socially sensitive contexts, such as displacement in Brooklyn neighborhoods. These precedents prove that machine learning methods, especially clustering and classification, are capable of predicting gentrification patterns with a certain degree of interpretability.

Displacement, however, is not sufficiently studied along with gentrification. Many scholars have defined displacement as a kind of forced residential mobility. One of them is Peter Marcuse, who considers displacement as a process that involves the direct or indirect forced removal of households from their homes as a consequence of gentrification (and abandonment) (Marcuse,

1985). He identified four kinds of displacement: Direct last-resident displacement, Direct chain displacement, Exclusionary displacement, and Displacement pressure (see Table 1). Yet Watt and Morris (2024) try to delink the two concepts and argue that displacement can also be caused by other urban processes apart from gentrification-related neighbourhood change. They admit that quantifying displacement (especially gentrification-induced displacement) is difficult and see it as a major methodological challenge for contemporary urban displacement research.

Displacement Type	Definition	Example
Direct Last-Resident Displacement	The number of the last resident of certain housing units affected	Can be physical (e.g. forced eviction) or economic (e.g. a rent increase)
Direct Chain Displacement	Households occupied certain units earlier and have been forced to move at an earlier stage in the physical decline of the building or an earlier rent increase	Can be physical (e.g. forced eviction) or economic (e.g. a rent increase)
Exclusionary Displacement	Households cannot access housing as it has been gentrified/abandoned and excluded from living where they would otherwise have lived.	“When one household vacates a housing unit voluntarily and that unit is then gentrified or abandoned so that another similar household is prevented from moving in, the number of units available to the second household in that housing market is reduced.” (Marcuse, 1985: p 206)
Displacement Pressure	The dispossession suffered by poor and working-class families during the transformation of the neighbourhoods where they live	“When a family sees the neighbourhood around it changing dramatically... all clearly are making the area less and less livable, then the pressure of displacement already is severe.” (Marcuse, 1985: p 207)

Table 1. Four Types of Displacement (Source: (Marcuse, 1985; Slater, 2009))

There are some attempts to study displacement and gentrification with a positivist approach, and New York City has become the forefront of these attempts, given its abundance of data. Freeman and Braconi (2004) wrote a provocative article that argues gentrification does not cause displacement directly. They use logistic regression models for the New York City Housing and Vacancy Survey (NYCHVS) data, and find that displacement, which is problematically indicated by disadvantaged households that moved, is not statistically associated with gentrification. Newman & Wyly (2006) replicate their methods while adding more important variables like rent-burdened households, and mixed quantitative results with in-depth interviews. They challenge Freeman and Braconi's findings and argue that NYCHVS data underestimates displacement. The reliance on family movement as a binary indicator of displacement in both research presents limitations. Carlson (2020) identifies three approaches in quantifying displacement, including the population approach, the individual-level approach, and the motivational approach. He tested the three approaches with NYCHVS data. In conclusion, he warned that all of these approaches only provide proxy measures and cannot completely capture the complex mobility behaviors, and a lack of consistent proportionality across approaches might lead to biased estimates of neighborhood displacement. Easton et al. (2019) review the research on quantifying displacement and highlight the overt research gap in this field, especially when taking Marcuse's typologies of displacement and the temporal dynamic of urban change into consideration.

To sum up, in the quantitative field, measuring displacement and unpacking its relationship with gentrification is still a major challenge. It is apparent that a multivariate measurement of displacement is crucial to the study of gentrification, or neighborhood change in general. To inform effective policy and mitigate the adverse effects of displacement, a deeper understanding of the socio-economic processes driving displacement is essential. Moreover, using methods that allow

for prediction could offer insights into how alternative urban planning strategies might impact outcomes.

Data and Method

“The full impact of displacement must include consideration of all four forms.... It must include displacement from economic changes, physical changes, neighbourhood changes, and individual unit changes.” (Marcuse, 1985: p. 208)

We aim to study displacement with a multi-dimensional measurement, including variables that can represent Marcuse’s four types of displacement. Instead of using displacement as a binary dummy variable in a statistical model, we also aim to put different places into distinct categories according to their displacement status. Machine learning offers us a delicate tool set to achieve such goals. Unsupervised learning, like clustering, can help identify distinct displacement patterns based on a multitude of variables, allowing for a more nuanced approach compared to traditional statistical logistic regression models. Supervised machine learning models excel in detecting patterns and generating predictions, offering a powerful tool for advancing this field. Techniques like SHAP can further shed light on the relative importance of different variables, helping to demystify the “black box” of displacement processes. Analysing census data at the census tract level can also reveal the spatial distribution pattern of displacement and gentrification, deepening our understanding of ongoing and future urban change.

Data Collection

In our research, in consideration of data availability and feasibility, we only study the economic-physical displacement of residents. Our dataset includes a variety of indicators (29 indicators) related to demographic, economic, and housing changes between 2014 and 2019. We get the ACS 5-year estimate census data in 2014 and 2019 for census tracts in Kings County, NY, with census api, and collect other data from NYC Open Data and the dataset of PLUTO/MapPLUTO by the Department of City Planning. They can be divided into broad categories including:

- 1) **Population Changes:** Changes in total population.
- 2) **Migration and Household Composition:** Changes in migration patterns and female-headed households.
- 3) **Socioeconomic Shifts:** Changes in the population below the poverty line, racial and ethnic composition (e.g., non-Hispanic white, Hispanic or Latino, Black or African American, and Asian) .
- 4) **Housing and Rent Dynamics:** Changes in median gross rent, owner-occupied and renter-occupied, rent burdened and mortgage.
- 5) **Employment and Economic Factors:** Changes in total employment, service and production occupation population and percentages, and changes in female service occupations, Changes in gentrifiers (gentrifiers are defined as Management, business, science, and arts occupations and Sales and office occupations)

We performed data cleaning by excluding census tracts with a total population of less than 500.

Clustering

We used a clustering model that has variables related to Peter Marcuse's four types of displacement, and get a robust result achieving similar results for k-means and hierarchical clustering. As key features, six variables were selected to make clustering analysis such as change in : 1) below poverty line 2) median gross rent 3) rent burdened 4) renter occupied 5) migration 6) gentrifiers. Each of them are directly relevant to Peter Marcuse's displacement type that is suggested above. For example, change in rent burdened and renter occupied are for direct resident-last displacement and direct chain displacement. In addition, we used MaxAbsScaler for both k-means and hierarchical cluster analysis.

To proceed with K-means clustering, elbow method was applied to consider the appropriate cluster size which was 5 in this case (Appendix Figure 1). It also displays cluster centers representing the mean values of the features for each cluster. Then with executed cluster distribution (Appendix Figure 2), the data was merged with census tract shapefile for further visualized spatial analysis. The table shows cluster center values for the selected variables. These centers represent the average values of the features within each cluster.

Classification

With the cluster analysis, we proceeded with classification model adding non-ACS data such as land use and affordable housing construction data looking at changes between 2014 and 2019. The land use data is loaded from PLUTO and MapPLUTO. Using ArcGIS, we identified and filtered census tracts that experienced change in land use categories. The filtered data of land use change and Affordable Housing construction data, focusing on total affordable housing units were merged with census tract data.

In Classification model, we ran several different models such as logistic regression, XG Boost, Random Forest and Multilayer Perceptron to test their accuracy of train-test split. In the below section, we will continue with the results and analysis of the models that showed the highest accuracy on test scores and meaningful implications.

Result and Analysis

First, we used the K-Means model to perform the clustering analysis. According to the elbow method, the silhouette method, and Marcuse's theory, we set the number of clusters to 5. The table below shows the means of our cluster centers. We interpret the result based on the means of cluster centers and identify certain patterns that align or resonate Marcuse's four types of displacement. Cluster 4, which lose the most population below poverty line, appeared to be dominated by the two types of direct displacement. Cluster 3, on the other hand, has a huge surge in median gross rent and influx of professionals, which might represent a dominant displacement pressure posed by neighborhood change. While Cluster 1 is not dramatically losing low-income population and only has a minor increase in rent, it has a significant loss of neighborhood mobility, which might imply exclusionary displacement instead of Freeman argument of succession (Slater, 2009; Freeman, 2005).

	Change_below_ poverty_line	Change_median_ gross_rent	Change_rent_ burdened	Change_renter_ occupied	Change_ migration	Change_ gentrifiers
Cluster 4	-0.378	0.244	-0.435	-0.356	-0.372	-0.100
Cluster 3	-0.127	0.481	-0.031	0.052	0.060	0.226
Cluster 1	-0.086	0.069	-0.153	-0.140	-0.159	-0.101
Cluster 0	0.052	0.025	0.103	0.090	0.099	0.054
Cluster 2	0.285	-0.054	0.456	0.390	0.386	0.167

Table 2. Cluster Centers for K-Means Clustering Result

To test the robustness of our analysis, we also conducted hierarchical clustering and got similar results to the k-means clustering. Below map is a map with the Neighborhood Tabulation Area (NTA) boundaries overlaid above the k-means clustering analysis, with clusters ranked by the change of population below poverty line from the lowest to the highest. The result shows that Downtown Brooklyn, Green Point, Williamsburg, Carroll Gardens-Cobble Hill-Gowanus-Red Hook, Park Slope, Windsor Terrace-South Slope, Prospect Height, Fort Hamilton, Coney Island-Sea Gate, Dyker Heights, Mapleton-Midwood (West), and Marine Park-Mill Basin-Bergen Beach are NTAs that have the most severe displacement. Compared with research that uses NYCHVS data (Carlson, 2020), Bushwick, Bedford-Stuyvesant, and Ocean Hill-Bronsville in our clustering analysis does not appear to be suffering from severe displacement.

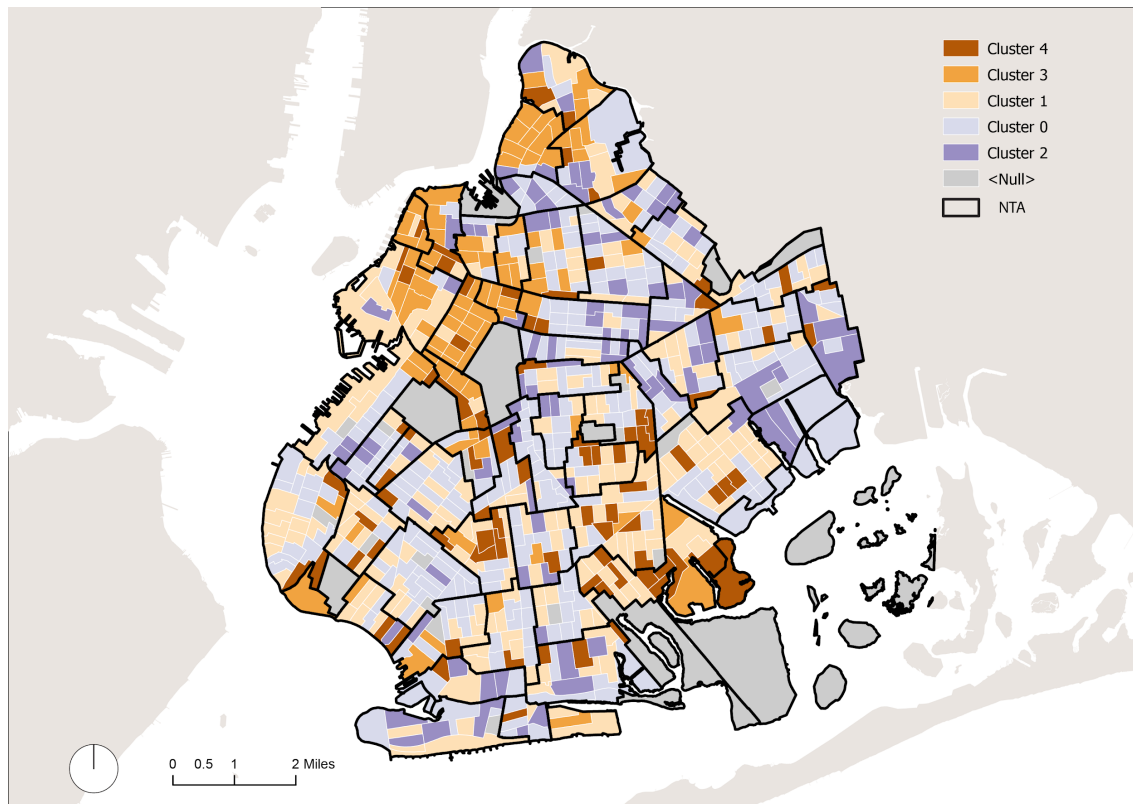


Figure 1. K-Means Clustering Result
(cluster ranked by change of population below poverty level)



Figure 1 (continued). Radar Plot Showing The Cluster Centers

The table below shows the comparison of multiple types of classification and their scores on multiple types of classification only with ACS data, and achieves high accuracy and precision ranging from 0.717 up to 0.891.

	Logistic Regression	Decision Tree	XG Boost	Multilayer Perceptron
Accuracy	0.804	0.717	0.826	0.891
Precision	0.766	0.709	0.830	0.844
Recall	0.679	0.701	0.826	0.836
F1 Score	0.658	0.702	0.821	0.834

Figure 2. Accuracy Test Score Comparison

SHAP Analysis For Supervised Learning

Logistic Regression SHAP Analysis

From the SHAP analysis summary for Logistic Regression model, the top features that have the largest spread of SHAP values having the strongest impact on the model output are: change in total housing units, median gross rent, overcrowding, black population and bachelor degree holders change.

This suggests that areas undergoing housing development, rising rents and demographic shifts toward more educated and white populations are more likely to be classified as gentrified. Areas with a decline in the Black population and an increase in the White population are strongly associated with gentrification (Class 4). The education level (bachelor's degree holders) also aligns with demographic shifts, implying the displacement of less educated, predominantly Black residents.

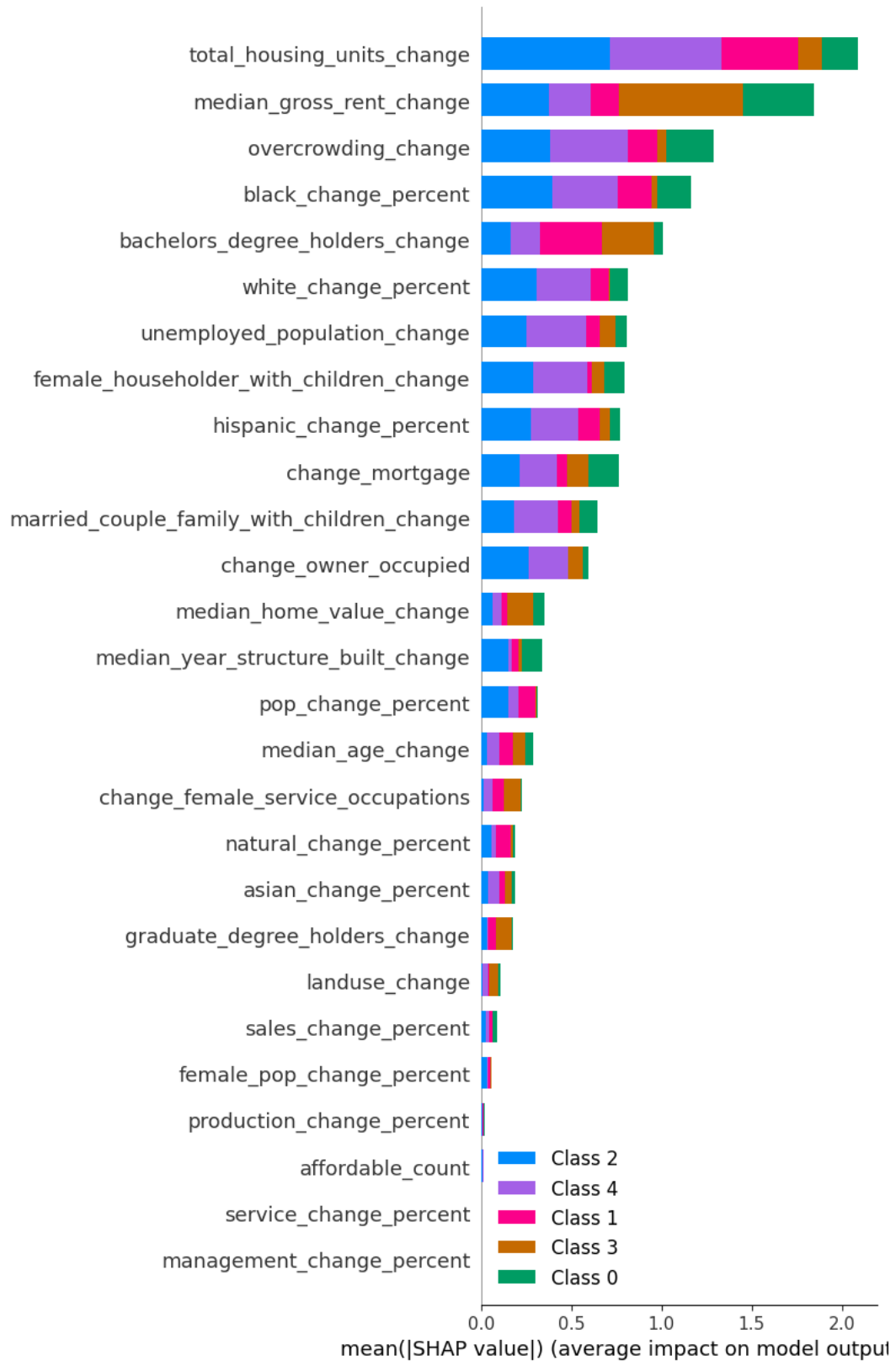


Figure 3. Logistic Regression SHAP Analysis

Decision Tree SHAP Analysis

The decision tree SHAP analysis (Figure 4) shows population change as the most influential indicator, followed by median gross rent changes and total housing unit changes. One thing to note here is that the change in the female householder with children becomes 4th important factor, indicating the change in the marginalized population, as such is a signal for gentrification .

XG Boost SHAP Analysis

From the XGBoost SHAP analysis (Figure 5), the most influential features on neighborhood classification are: changes in median rent, total housing units, population, and ownership patterns. The model indicates that gentrification is driven by rising housing costs, new developments, and demographic shifts, particularly in the female population and homeownership rates. Similar to SHAP analysis of logistic regression, median gross rent change and total housing units change take an important place as indicators of gentrification.

Multilayer Perceptron SHAP analysis

The SHAP analysis from the Multilayer Perceptron (MLP) model result is similar to Logistic Regression SHAP analysis. MLP also indicates that total housing units change, median rent change, and changes in owner-occupied housing are the most influential factors in predicting neighborhood transformation. The model shows that gentrification is driven by new housing developments, rent increases, and demographic shifts, particularly the influx of more educated, white populations.

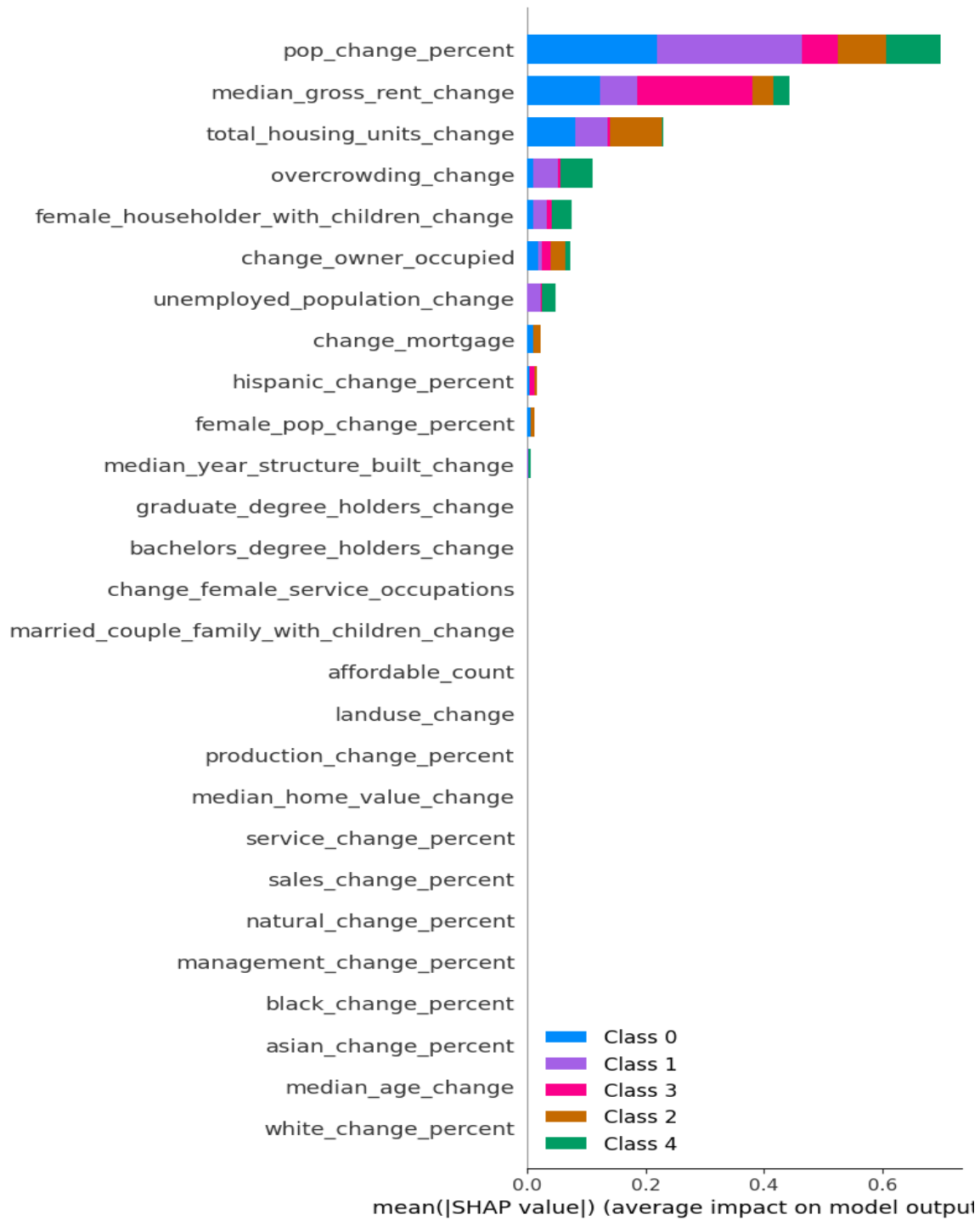


Figure 4. Decision Tree SHAP Analysis

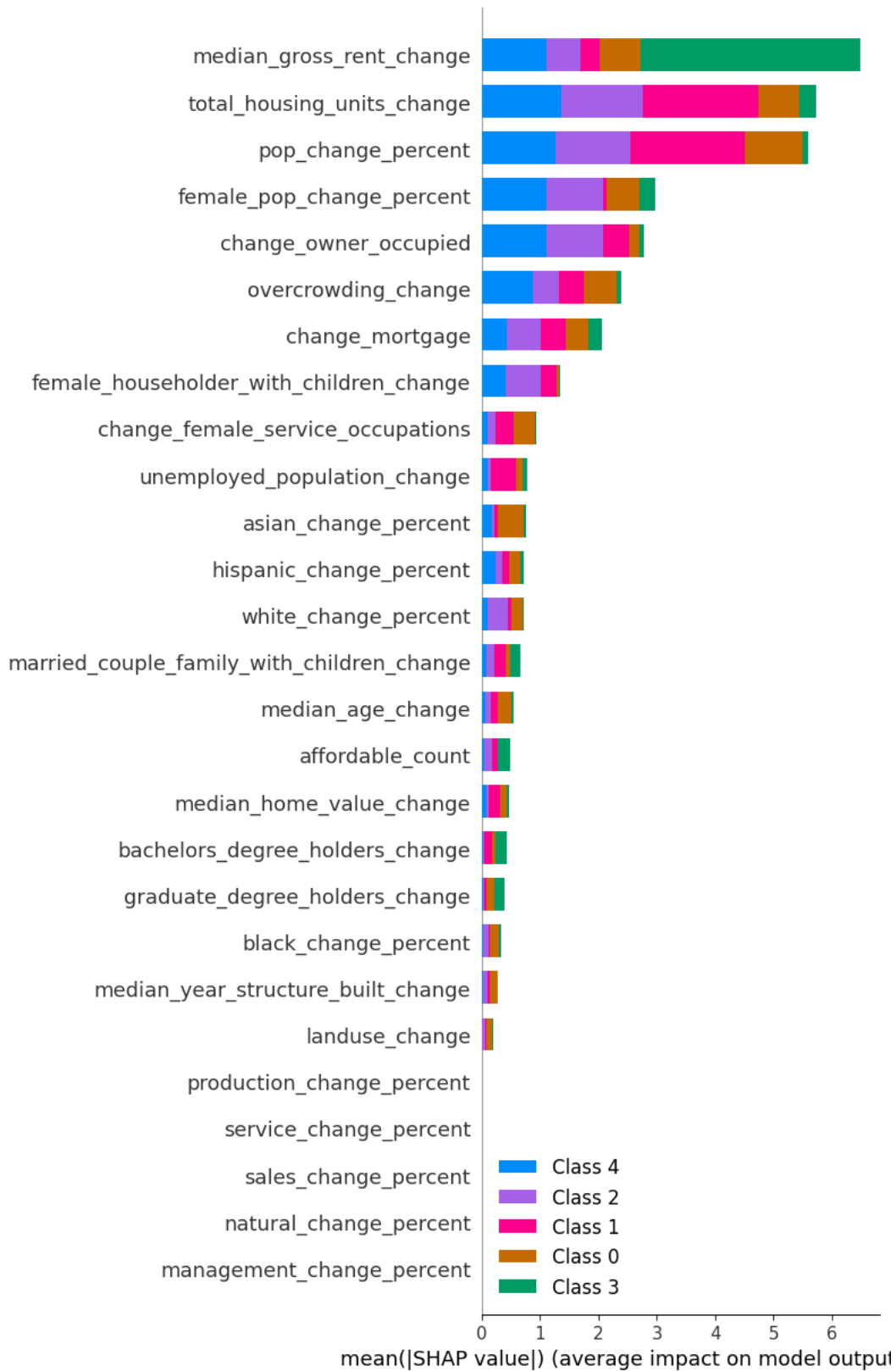


Figure 5. Decision Tree SHAP Analysis

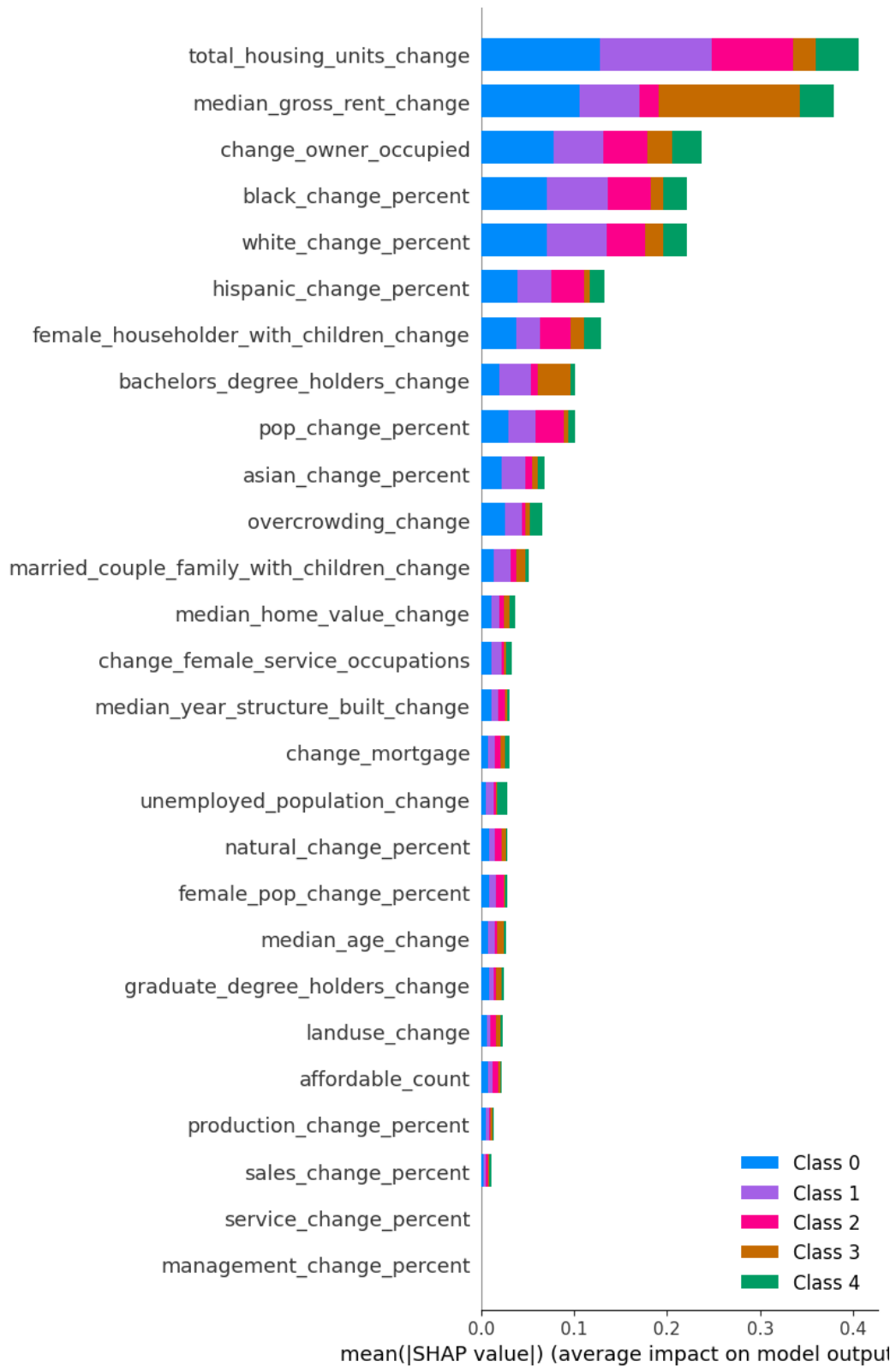


Figure 6. Multilayer Perceptron SHAP analysis

Synthesis

Rank	Logistic Regression	Decision Tree	XGBoost	MLP
1	Total Housing Units Change	Total Population Change	Median Gross Rent Change	Total Housing Units Change
2	Median Gross Rent Change	Median Gross Rent Change	Total Housing Units Change	Median Gross Rent Change
3	Overcrowding Change	Total Housing Units Change	Total Population change	Owner Occupied Units Change
4	Black Population Percentage Change	Overcrowding Change	Female Population Change	Black Population Percentage Change
5	Bachelor's Degree Holders Change	Female Housholder with Children Change	Owner Occupied Units Change	White Population Percentage Change

Figure 7. Summary of Top 5 SHAP Values for different models

The SHAP analysis from four different models—Logistic Regression, Decision Tree, XGBoost, and Multilayer Perceptron (MLP)—provides a comprehensive understanding of gentrification and neighborhood transformation. Despite differences in model architecture, there are consistent patterns that emerge across the analyses. To synthesize SHAP analysis of four models, we can identify common findings across models, which are total housing units change, median gross rent change, and demographic changes such as race (white and black population), and education increase in bachelor's degrees.

The SHAP analysis from all four models (Figure 7) reveals that displacement is driven by new housing developments (demolitions and construction), rising rents, and demographic shifts, particularly toward more educated and white populations, which indicates gentrification. While

the Logistic Regression and MLP models emphasize the role of education and racial change, the Decision Tree model uniquely highlights the vulnerability of marginalized populations like female-headed households. Meanwhile, the XGBoost model highlights homeownership transitions as a significant factor. However, the analysis also draws attention to subtle and smaller waves of displacement, which may indicate the early signs of neighborhood change.

Conclusion and Discussion

This article has shown how machine learning can allow for a multivariate measurement of displacement. Following Marcuse's conceptualization, we trace potential factors that can reveal direct last-resident displacement, direct chain displacement, exclusionary displacement, and displacement pressure. We find that on the census tract level, clustering can identify different displacement patterns in Brooklyn that empirically reflects the typologies. Our clustering result also highlights the importance to study indirect displacement.

With supervised classification methods, we explored how displacement is interconnected with gentrification, and to what extent do social, economic, demographic, and physical factors impact neighborhood displacement patterns. The SHAP analysis across all four models — Logistic Regression, Decision Tree, XGBoost, and Multilayer Perceptron — demonstrates a clear and consistent narrative: displacement is primarily driven by structural factors like housing unit changes, rising rental costs, demographic shifts, and the intersection between class, race, and gender. Notably, while the Logistic Regression and MLP models underscore the impact of racial and educational changes, the Decision Tree model draws attention to shifts in vulnerable household types, such as female-headed families. XGBoost, meanwhile, brings a unique

perspective by emphasizing ownership dynamics. The convergence of insights from diverse models strengthens the argument that indicators of gentrification can also be used to predict displacement.

We assume land use change and the construction of affordable housing would be important factors in predicting displacement. However, they never enter the top ten important factors in the SHAP plot. This could imply that change in physical environment does not directly affect displacement, and the claim that building affordable housing (like in the Low Income Housing Tax Credit program) can effectively mitigate displacement (Freeman & Braconi, 2004) needs further verification.

Our research carries several important implications for both urban theory and policy. First, our findings empirically support Marcuse's framework on displacement, particularly his distinction between direct displacement, exclusionary displacement, and displacement pressure. Through our machine learning-driven approach, we were able to validate how these different forms of displacement represented spatially and demographically in Brooklyn neighborhoods, reinforcing the relevance of his typology in contemporary contexts.

Second, our clustering analysis allowed us to identify specific census tracts and neighborhoods that are experiencing the most severe levels of displacement. These areas often share common characteristics such as rising housing costs, demographic turnover, and declining shares of low-income or historically marginalized populations, which can be monitored as priority zones for intervention by related agencies.

Third, our classification models reveal a critical insight: the construction of affordable housing, while necessary, is insufficient on its own to mitigate displacement. Without complementary measures such as stronger tenant protections, rent control, and anti-displacement

policies, affordable housing may be undermined by market pressures and fail to reach those most at risk of displacement. Together, these implications underscore the need for a multidimensional approach to urban planning that integrates predictive analytics with community-centered strategies.

This research sheds light on the opportunities of using census tract-based data (e.g., ACS) in evaluating and addressing displacement and gentrification through machine learning. By incorporating non-ACS data, such as affordable housing data, it highlights the need to reevaluate current policies in terms of reconciling the conflicts between displacement and gentrification. With ACS data being georeferenced, it opens up further exploration of spatial analysis, which is effective in identifying neighborhoods where these conflicts are happening and more. As demonstrated by the SHAP analysis from four different models, there is a need for a holistic approach to policy and mitigation initiatives aimed at addressing displacement and gentrification, not only in Brooklyn but also in other cities facing similar challenges.

However, our machine learning approach and research process have some limitations. First, the features in our clustering model are intentionally chosen according to Marcuse's theory, which might omit displacement that are not caused by gentrification. Due to considerations of feasibility, we only applied socio-economic indicators, without regarding temporality, vulnerability to displacement, and its emotional impacts (Carlson, 2020).

Second, unlike statistical models, classification models does not allows for testing p-values and measuring interactions between variables. Compared with previous research, our approach suffers interpretability in the trade-off of accuracy and the capacity to measure displacement with higher dimension. We also only used cross-sectional data in our research, which lacks the explanation of temporal shifts like the longitudinal logistic models did.

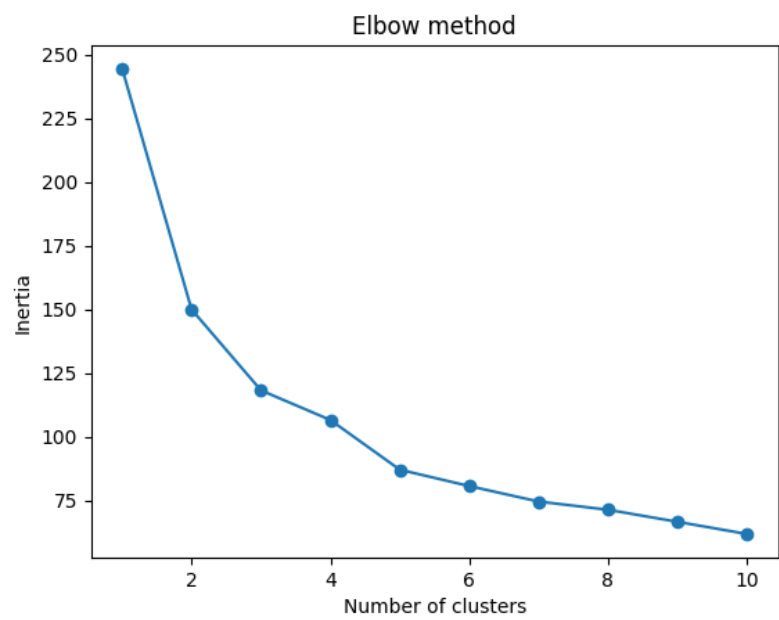
Third, as Marcuse said, it is difficult to measure displacement at the census tract level directly with census data (Marcuse, 1985). As Carlson (2020) said, our clustering model only provide a proxy measurement to the intricate process of urban displacement in the real world. It is also important to verify the results in this paper with in-depth interviews and participatory action research. Overall, measuring and predicting displacement is a challenging task, yet machine learning offers a promising alternative approach to disentangle the black box of displacement and its root in gentrification.

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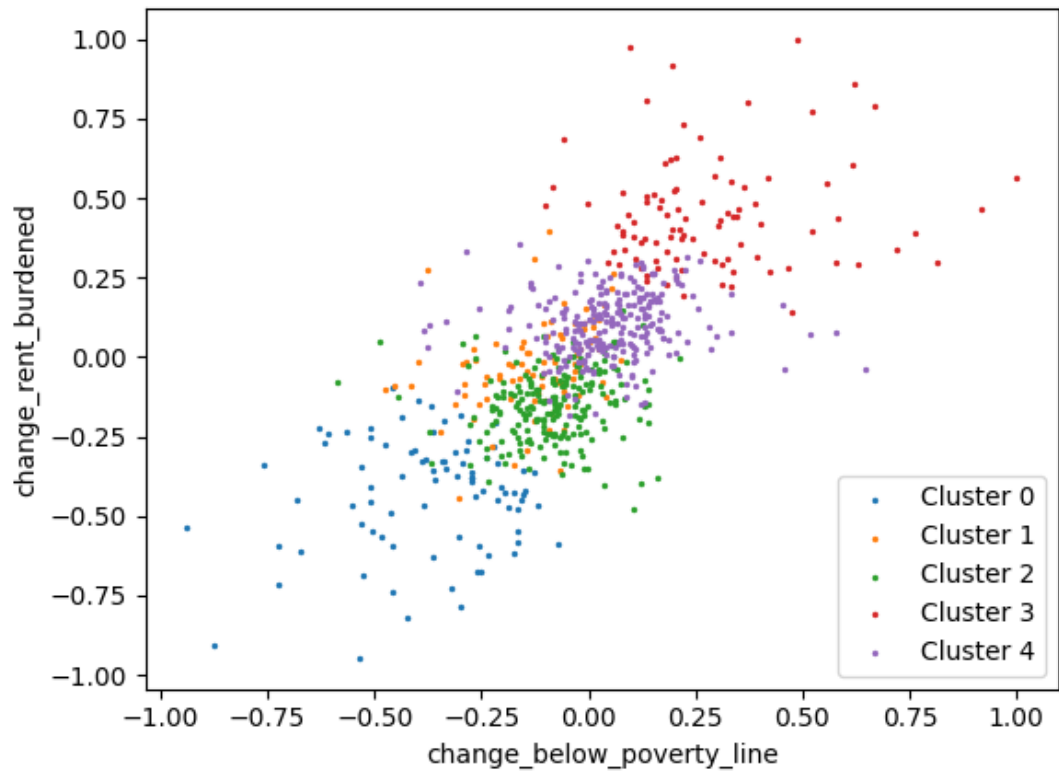
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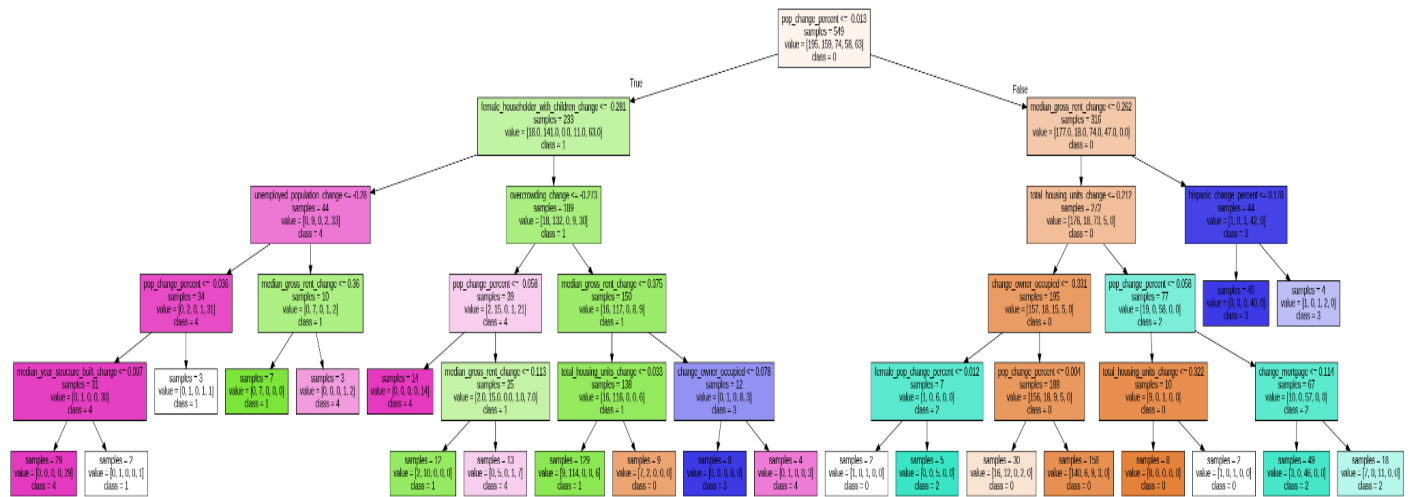
Appendix



Appendix 1. Elbow Method



Appendix 2. Cluster Distribution



Appendix 3. Decision Tree